

Vaibhav Kumar Mahawar

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Portfolio | Ajmer, Rajasthan, India

Education

Dr. B. R. Ambedkar National Institute of Technology Jalandhar

Master of Technology in Machine Intelligence and Automation CGPA: 8.48/10

Jalandhar, Punjab

Aug. 2024 – Jun. 2026

Poornima Institute of Engineering and Technology

Bachelor of Technology in Computer Science and Engineering CGPA: 8.54/10

Jaipur, Rajasthan

Jul. 2019 – Jun. 2023

Experience

Software Development Intern

Jul. 2022 – Sep. 2022

Techienest Pvt. Ltd.

Jaipur, Rajasthan

- Developed a React Native mobile application for 500+ primary students featuring interactive Vedic mathematics exercises, progress tracking, and REST API integration, improving learning accessibility across Android devices.
- Collaborated with cross-functional teams to develop, test, and optimize scalable mobile application features while integrating REST APIs and improving application performance and maintainability.

Projects

RudraAI– AI-Powered Software Engineering Platform | Node.js, Typescript, Python, Redis, Neo4j, Docker 2026

- Architected an autonomous multi-agent software engineering platform capable of planning, code generation, repository analysis, debugging, documentation synthesis, and deployment using FastAPI, MCP, and LLM orchestration.
- Designed scalable backend services with FastAPI, Redis, and Neo4j while implementing memory-enabled reasoning, tool execution, and intelligent agent collaboration for software engineering workflows.

Career Twin– AI Career Intelligence Platform | React, Typescript, AWS Bedrock, LangGraph, Vector Search 2026

- Developed an AI-driven career intelligence platform that analyzes resumes, matches job descriptions, identifies skill gaps, and generates ATS-optimized resumes, cover letters, and interview preparation plans.
- Implemented semantic retrieval, Retrieval-Augmented Generation (RAG), vector search, and AWS Bedrock-powered reasoning pipelines to deliver personalized career recommendations through scalable backend services.

AI-Powered Drone Intelligence Platform | Python, YOLOv8, OpenCV, LangChain, LLMs, ANFIS, RAG 2026

- Developed an AI-powered drone intelligence platform integrating YOLOv8, OpenCV, and ANFIS for real-time drone detection, threat assessment, and explainable decision support across surveillance scenarios.
- Integrated Retrieval-Augmented Generation (RAG) with LLMs to enable natural language querying, surveillance summarization, contextual reasoning, and explainable analysis of detected aerial threats.

Research Experience

Real-Time Drone Detection and Classification System | Python, YOLOv8, OpenCV, ANFIS, Fuzzy Logic 2026

- Developed a real-time YOLOv8-based drone detection framework with ANFIS-enhanced classification, achieving over 91% detection accuracy while reducing bird–drone misclassification by 52%.

Generative AI and Agentic Systems Research | LLMs, LangChain, LangGraph, MCP, RAG, AI Agents Ongoing

- Exploring autonomous AI agents, Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP), long-context reasoning, and multi-agent orchestration for enterprise-scale AI applications.

Technical Skills

Programming Languages: Python, Java, C++, JavaScript, SQL

AI & Machine Learning: PyTorch, TensorFlow, Scikit-learn, OpenCV, YOLO, Computer Vision, CNNs, Transformers, Vision Transformers, Transfer Learning, ANFIS, Fuzzy Logic

Generative AI: OpenAI, Gemini, LangChain, LangGraph, LlamaIndex, Prompt Engineering, RAG, MCP, AI Agents, Multi-Agent Systems, Tool Calling, Embeddings, Vector Search

Backend Development: FastAPI, REST APIs, Node.js, Express.js

Databases: Neo4j, Redis, PostgreSQL, MongoDB, SQLite, FAISS, Weaviate

Cloud: Azure AI, AWS, Docker, Kubernetes, Git, GitHub, GitHub Actions, Linux

Data Engineering: Pandas, NumPy, PySpark, ETL Pipelines, Data Processing, Feature Engineering

Tools: VS Code, Eclipse, Jupyter Notebook, Postman, Linux

Certifications & Achievements

Research Publications Published research papers in international journals and conferences on Computer Vision, Artificial Intelligence, Intelligent Surveillance, Photovoltaic Systems, and Renewable Energy Optimization.

Microsoft Certified: Azure AI Fundamentals (AI-900) Certified in Azure AI services, Machine Learning fundamentals, Responsible AI, and AI solution development.

GATE Qualified Qualified GATE Computer Science (2024 & 2026) and GATE Data Science & Artificial Intelligence (DA 2026).